University of Roehampton London

ASSESSMENT BRIEF (Part 1) (Draft)

<table>
<tr><td>Programme</td><td colspan="3">MSc Artificial Intelligence</td></tr>
<tr><td>Module Title</td><td colspan="3">AI Systems Engineering</td></tr>
<tr><td>Module Code</td><td colspan="3">CMP-L044</td></tr>
<tr><td>Module Level</td><td>7</td><td>Assessment Type(s)</td><td>Report</td></tr>
<tr><td>Word Length</td><td>2000</td><td>Contribution to Module Mark</td><td>40%</td></tr>
<tr><td>Submission Deadline (Date & Time)</td><td>14:00
8 April 26</td><td>Submission Format</td><td>PDF/DOCX</td></tr>
<tr><td>Feedback Date</td><td>Within a month of submission</td><td>Submission Location</td><td>Moodle</td></tr>
<tr><td colspan="2">Learning Outcomes
This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the learning outcomes listed below.
Critically evaluate the principles, methodologies, and frameworks for designing, developing, and deploying scalable, reliable, and reproducible AI systems across diverse domains.
Design and implement comprehensive AI pipelines, encompassing data engineering, feature engineering, model development, and deployment strategies, using tools and frameworks adopted in the industry
Assess the performance, scalability, and monitoring needs of AI systems, employing advanced techniques to ensure effective operation in production environments.
Apply advanced engineering principles to develop AI solutions for real-world scenarios, addressing domain-specific</td><td colspan="2">Employability and Professional Skills
This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the following skills, which are critical in a professional context and in jobs.

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This assessment develops employability and professional skills aligned to AI engineering. Working as part of an engineering team, you will practise requirements elicitation, system-level design, and evidence-based decision-making. You will produce a clear, auditable design proposal that integrates data, model, deployment, monitoring, and governance considerations, mirroring professional design reviews.</td></tr>
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challenges, including reliability, robustness, and ethical implications. • Integrate and apply principles of Responsible AI, including fairness, transparency, and compliance with data privacy regulations, to design ethical AI systems in production environments, ensuring alignment with Responsible AI standards.	

Assessment Requirements

Part 1: Group Critical Appraisal and Proposal (40%)

Overview:
Part 1 of the summative assessment for this module is a group-written report that accounts for 40% of the final mark.

Purpose
To demonstrate systems-level thinking and critical appraisal by defining a realistic AI system problem and proposing an end-to-end engineering solution that is scalable, reliable, reproducible, and responsibly governed.

Description

Propose an AI application with a clearly articulated context. Define the system boundary, stakeholders, operating environment, and success criteria. Critically appraise at least two relevant AI systems engineering approaches or design options (e.g., architectural styles, pipeline patterns, deployment strategies, or monitoring approaches). The critical appraisal must be supported by recent peer-reviewed research papers.

You should evaluate the contributions of these studies, analyse their strengths and limitations, identify gaps or unresolved challenges, and justify trade-offs. Based on this analysis, propose a justified extension, improvement, or alternative design approach that addresses the identified limitations in existing work.

This expectation reflects current research practice in AI systems engineering, where modern AI systems are treated as end-to-end infrastructures involving data pipelines, model training, deployment, and monitoring components, rather than isolated algorithms.

Your proposal must include, as appropriate:

- Problem statement and motivation, presenting a clearly articulated AI application context, including key constraints, risks, and the intended success criteria for the system.

- System context and boundary, identifying stakeholders, the operating environment, and interactions with external systems, supported by a context diagram or equivalent representation.
- High-level system requirements, including clearly defined functional and non-functional requirements aligned with the problem definition and system objectives.
- Critical appraisal of existing approaches, supported by recent research papers, evaluating at least two relevant AI systems engineering approaches or design options (e.g., architectural styles, pipeline patterns, deployment strategies, or monitoring approaches). The analysis should assess contributions, identify strengths and limitations, highlight gaps, and justify design trade-offs.
- Proposed AI system architecture and data strategy, describing the end-to-end system design, including data pipelines, model components, serving infrastructure, and the data and experimentation strategy (data sources, versioning, provenance, and evaluation design).
- Deployment and operations considerations, outlining how the system would be deployed and managed in practice, including reproducibility, monitoring, and incident response mechanisms.

Declaration of Use of Generative AI

You must explicitly declare any use of generative AI tools in the preparation of this assessment. This includes, but is not limited to, tools such as ChatGPT, GitHub Copilot, Google Gemini, or other text generation, code generation, summarisation, or paraphrasing systems.

If generative AI tools were used, you must clearly state:
• The name of the tool used and the purpose for which it was used (e.g., idea generation, language refinement, code suggestions).
• The specific sections of the work where the tool contributed.

This declaration should appear in a short statement at the beginning of your report.

If no generative AI tools were used, you must include a clear statement at the beginning of the document confirming that no generative AI tools were used in the preparation of the submission.

Students remain fully responsible for the accuracy, originality, and integrity of all submitted work. The use of generative AI tools must comply with the university's academic integrity regulations, and any undeclared use may be treated as academic misconduct.

Deliverable

Submit a single 2,000-word report (excluding references and appendices) using the Part 1 Report Submission Template available on Moodle. Your report must include an individual contribution statement (brief bullet list per member in the Appendix).

File name format: use your group number only — e.g. Group_1.*pdf*.
The report should:

- Present a clearly structured argument with logical flow between sections.
- Include a comprehensive literature review that contextualises the chosen problem.
- Demonstrate academic rigour and clarity of expression appropriate to Level 7 study.
- Include accurate citations and a reference list in IEEE format.
- Conclude with a summary of key insights and identified limitations.
- There is no ±10% leeway on the stated word count.

Marking Criteria

See Table 1: Assessment Criteria

Assessment Success Guidance

High-performing submissions demonstrate rigorous critical appraisal, clear requirements and architecture rationale, and a coherent end-to-end design that is feasible in practice. You should explicitly justify trade-offs (e.g., latency vs cost, accuracy vs interpretability, batch vs streaming, cloud vs edge) and show how reproducibility and monitoring are engineered from the outset. Responsible AI should be integrated into the design (not added as an afterthought), with concrete mitigations and governance mechanisms. Writing should be concise, precise, and reflective, demonstrating an understanding of the ethical and sustainability implications while maintaining IEEE referencing and a professional presentation.

Assessment Guidance, Support, and Formative Feedback

Students are continually provided with opportunities to seek clarification and receive guidance on the assessment throughout the module. During scheduled sessions, they can present draft work, discuss their problems and proposed approaches, and obtain formative feedback to improve their final submission. In addition, Week 12 is dedicated to an Assessment Workshop, where students can consolidate their understanding of the assessment criteria, refine their work, and address any remaining questions before submitting their final work.

Contact for Queries

Module Leader: Dr Mohammad Javaheri

Use of Artificial Intelligence (AI)

*The assessment is designed so that the use of AI during the assessment is **possible**. You must acknowledge any use of AI and appropriately cite all AI-generated outputs. Please ensure you read and understand the assessment guidelines and consult your Module Leader if you have any questions. Guidance on AI use:*

- *University AI Principles and Guidance for Senate*
- *Student Guidelines on the Use of AI*

Referencing

IEEE Referencing Style

Mitigating circumstances/late penalties

Sometimes, circumstances outside of your control may affect your studies and might prevent you from submitting work on time or attending an exam.

The University offers the ability for students to request additional time to complete an assessment or to defer an examination to a later date. If you are finding yourself in such a situation, please speak to your Academic Guidance Tutor, the Roehampton Student Union (RSU) or someone in the Wellbeing team first, who can support you. Further details are available on the mitigating circumstances portal.

If you do not apply for or are not approved for Mitigating Circumstances, late penalties will apply. If work is submitted up to 14 calendar days late, the mark will be capped at 50%; if it is over 14 calendar days late, it will not be marked.

Resubmissions and Reassessment

If you are required to resubmit this assessment or take part in reassessment, you will be notified via Moodle and your student email. Please ensure you check both regularly. Any reassessment tasks will follow the same learning outcomes and criteria.

Submission Checklist

Before you submit, ask yourself:

Have I fully answered the assessment brief?

Have I met the word count and formatting requirements?

Is my referencing complete and accurate?

Have I declared any AI use honestly?

Have I proofread my work?

Am I submitting through the correct platform before the deadline?

Table 1: Assessment Criteria

Criteria	0-49% (Fail)	50-59% (Pass)	60-69%	70-79%	80-100%
Categorical Marks	25,35,38,42,45,48	52,55,58	62,65,68	72,75,78	82,85,92
System problem definition and context (10 Marks)	Unclear or incorrect problem definition; weak or missing system boundary and stakeholder context; minimal engagement with constraints and domain evidence.	Basic problem definition with limited context; boundary and stakeholders mentioned but weakly justified; limited consideration of constraints and risks.	Clear and relevant problem definition; system boundary and context articulated; constraints and risks identified with adequate justification.	Strong, well-justified problem definition; coherent boundary, stakeholders, and operational context; convincing constraints, risks, and success measures.	Exceptional clarity and rigour; comprehensive context and boundary reasoning; strong evidence base; nuanced treatment of constraints, risks, and success criteria.
Critical appraisal of approaches and trade-offs **(20 Marks)**	Minimal or incorrect discussion of chosen techniques; no theoretical depth or critical comparison; unclear understanding of underlying biological analogies or computational models.	Descriptive comparison with limited critical insight; some relevant concepts are discussed but lack integration or depth; inaccuracies in the theoretical explanation.	Competent comparative discussion; demonstrates sound understanding of algorithms and biological analogies; some synthesis across paradigms is evident.	Strong theoretical understanding; critical comparative analysis with good synthesis of ideas; well-referenced discussion showing awareness of algorithmic strengths, limitations, and assumptions.	Exceptional theoretical insight; rigorous, well-integrated comparative analysis showing synthesis across paradigms; supported by authoritative, recent literature; demonstrates intellectual originality.
Critical Evaluation and Proposed Design	Little or no critical appraisal; largely descriptive;	Some appraisal but limited criticality; trade-offs stated but weakly	Competent appraisal comparing design options; trade-offs	Rigorous, well-structured appraisal; trade-offs justified with strong	Outstanding critical insight and synthesis; trade-offs are argued with

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Categorical Marks	25,35,38,42,45,48	52,55,58	62,65,68	72,75,78	82,85,92
(30 Marks)	trade-offs absent or incorrect; weak evidence and limited understanding of AI systems engineering frameworks.	supported; limited integration of frameworks, standards, or relevant literature.	mostly justified with appropriate sources; demonstrates sound understanding of key frameworks and practices.	evidence and systems reasoning; integrates frameworks/standards and shows clear synthesis.	authoritative evidence; demonstrates originality in evaluation and strong professional judgement.
Proposed end-to-end design (architecture, pipeline, deployment, monitoring) **(30 Marks)**	Design is incomplete, infeasible, or incoherent; key components (data, model, serving, monitoring) are missing; reproducibility and operational considerations are absent.	Design covers some components but lacks cohesion; limited feasibility detail; weak reproducibility detail; the monitoring and validation plan is underdeveloped.	Coherent end-to-end design covering core components; reasonable feasibility; includes reproducibility and basic monitoring/validation plans.	Strong, feasible design with clear architecture and pipeline; well-articulated deployment and operations plan; robust reproducibility and monitoring considerations	Exceptional, production-aware design; comprehensive architecture and lifecycle plan; excellent reproducibility, monitoring, and incident response; strong validation strategy.

Criteria	0-49% (Fail)	50-59% (Pass)	60-69%	70-79%	80-100%
Categorical Marks	25,35,38,42,45,48	52,55,58	62,65,68	72,75,78	82,85,92
Academic Writing, Referencing and Responsible AI (10 Marks)	Poor structure and clarity; limited referencing or incorrect IEEE style; Responsible AI considerations absent or superficial.	Adequate structure but uneven clarity; referencing is inconsistent; Responsible AI is mentioned but lacks concrete mitigations or governance mechanisms.	Well-structured report with clear writing; mostly correct IEEE referencing; Responsible AI considered with some concrete measures.	Highly professional writing and presentation; accurate IEEE referencing; Responsible AI integrated with clear mitigations (fairness, privacy, transparency, security) and governance.	Exemplary, persuasive engineering report; flawless IEEE referencing; Responsible AI deeply integrated with strong, specific mitigations and governance aligned to standards and compliance expectations.